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ADAPTIVE POLITICAL ECONOMY
Toward a New Paradigm

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ABSTRACT

The conventional paradigm in political economy routinely treats living, complex, adaptive social systems as machine-like objects. This treatment has driven political economists to oversimplify big, complex social processes using mechanical models, or to ignore them altogether. In development, this has led to theoretical dead ends, trivial agendas, or failed public policies. This article proposes an alternative paradigm: adaptive political economy. It recognizes that social systems are complex, not complicated; complexity can be ordered, not messy; and social scientists should be developing the concepts, methods, and theories to illuminate the order of complexity, rather than oversimplifying it. The author illustrates one application of adaptive political economy by mapping the coevolution of economic and institutional change. This approach yields fresh, important conclusions that mechanical, linear models of development have missed, including that market-building institutions look and function differently from market-sustaining ones.

"ECONOMISTS tend to think in machine mode," Esther Duflo, who was awarded the Nobel Prize in 2019 for her pioneering work in randomized controlled experiments, observes. Her cowinner, Abhijit Banerjee also invokes the metaphor of machines to characterize the mission of economists: "Quesnay probably did not realize that when he . . . wrote in 1763 about what 'propels the economic machine,' he was launching what would become the dominant metaphor in economics." In unmistakably mechanical language, they call upon economists to function as the plumbers of economies: scrutinize the design of taps and the layout of pipes.[1]

These quotes by Duflo and Banerjee illustrate a habitual conceptual error in political economy: living, complex, adaptive social systems are routinely treated as machine-like objects. This approach misperceives the defining, nonmechanical characteristics of social systems and applies overly reductionist methods to studying and managing them. Put

[1] Duflo 2017, 2, 12.

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differently, it is a basic classification error, akin to treating liquids as solids, or more colorfully put, mistaking trees (complex, adaptive) for toasters (complicated, mechanical). Yet adherents of the mechanical worldview are often proud: they think that by treating trees as toasters, they simplify and control a messy world. In reality, as James Scott reminds us, treating nature as industrial objects has spawned a range of problems in real life, from ecological degradation to new diseases.[2]

Humans are facing a world of disruptions and radical uncertainty, where modernization—long celebrated uncritically as progress—is running up against its limits, the results of which include climate change, rising inequality, and geopolitical ruptures. Yet the classical, cherished paradigm—that is, the lens through which we view and make sense of the world—is inherited from the industrial age, a time of domination when humans revered machines and factories and sought to bring wild, messy nature under control. In short, we are using a mechanical paradigm passed down from the eighteenth through twentieth centuries to grasp the crises and possibilities of the twenty-first century. What is worse, we are doubling down on the paradigm of the past.[3]

What is the alternative? Social scientists should take inspiration from a pathbreaking approach in geometry, called fractals: the math of rough, complex shapes. These are found everywhere in nature, for instance, the veins on a leaf, the jagged relief of a mountain, or the curves of a face.

Fractals were discovered by Benoit Mandelbrot (at the age of 75, he was the oldest professor to receive tenure at Yale University), who spent the first thirty-five years of his career outside academia, at IBM. His work was initially rejected in every academic discipline, particularly by mathematicians who dismissed roughness in nature as messiness. They insisted that proper math should study only straight lines and even shapes within the grasp of existing formulas.[4]

Mandelbrot showed that roughness was, in fact, not senseless and chaotic, but surprisingly ordered. By the time he reached his seventies, his insights were transforming fields as diverse as biology, chemistry, engineering, medicine, physics, and even art. Life-like animations we

[2] Scott 1998.

[3] To be clear, my critique is not about randomized controlled experiments (RCTs). RCTs are a method, and advancements in any method should be applauded. My critique is about the paradigm underlying the dominance of RCTs in political economy, i.e., foundational assumptions about the nature of the social world. To RCT fundamentalists, societies are like machines. Granted, some aspects of societies may be mechanical, just as we sometimes find straight lines and even shapes in nature, but for the most part, societies are unlike machines.

[4] NOVA 2008.

enjoy watching today, for example, are thanks to the new math that he introduced.

We need the equivalent of fractals in social science. Instead of pretending that the uneven contours of social systems are straight lines, or ignoring them altogether, political economists should be developing the concepts, methods, and theories to illuminate the order of complexity.

The Math of Messy Shapes

My backyard is a little jungle of unkempt weeds, overgrown bushes, squirrels, groundhogs, and deer. Initially, this wilderness disturbed my instincts as an urban native who grew up in a big city decorated at regular intervals by manicured landscapes and shopping malls. But the three years that I was stuck with this jungle during the pandemic taught me something different. Staring up at the walnut trees, I noticed that the veins of a single leaf mirror the branches of a tree. You do not need to be a mathematician to sense a certain pattern, as if nature grows according to a formula.

What I intuitively perceived but could not name at the time are fractals—the geometry of rough, irregular shapes—discovered by Mandelbrot, a self-described "scientific maverick" and "outsider, moving from field to field."[5] The idea behind fractals is paradoxical: unorderly shapes can be highly ordered. Indeed, this order is so mathematically precise that it can be replicated in computer programs to create extremely realistic graphics. Although Mandelbrot's work is now celebrated as revolutionary, his ideas were scorned and rejected for decades because fractals "put him at odds with centuries of mathematical tradition."[6] As one mathematician explained:[7]

> We became used to the fact that certain patterns were amendable to mathematics: the architectural ones—largely the patterns of human-made structures where we had straight lines and circles and perfect geometric shapes. The basic assumption that underlines classical mathematics is that everything is extremely regular. I mean, you reduce everything to straight lines, flat surfaces. Classical mathematics is only well-suited to study the world that we've created, the things we've built . . . The patterns in nature . . . thus were outside of mathematics.

When I heard this comment, I felt a flicker of familiarity. Where had I encountered this logic before? In my own field of study, of course: the political economy of development.

[5] Mandelbrot 2013.
[6] NOVA 2008. Quoted from narration.
[7] NOVA 2008. Quoted from Keith Devlin, emeritus mathematician at Stanford University.

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Development analysts have long struggled with a chicken-and-egg problem. Which comes first in development, growth or good institutions?[8] In the temple of political economy, authoritative economists and political scientists have occupied different pedestals on this question.

One school, known as modernization theory, asserts that economic growth precedes and will drive political and institutional development, specifically democracy, but it cannot explain how an economy takes off in the first place.[9] For Jeffrey Sachs, the answer was to inject massive foreign aid with comprehensive, elite-designed programs.[10] In reality, however, his grand scheme backfired. Plans made in New York City were constantly sunk by unexpected problems on the ground.[11]

A second school, institutional economics, argues the reverse: good, modern institutions are necessary for economic prosperity.[12] But this school of thought cannot explain how a poor society that lacks capacity can quickly establish the ideal institutions needed for growth. In reality, decades of best-practice reforms led by international institutions to make poor countries "skip straight to Weber" have proven disappointing and even backfired. As Lant Pritchett and Michael Woolcock lament, it has been "a root cause of the deep problems encountered by developing countries."[13]

A third school, historical institutionalism, cleverly gets around the chicken-and-egg problem by underscoring the self-reinforcing effects of history. It argues that colonies inhabited by European migrants who brought inclusive, nonextractive institutions thrived, while colonies in Latin America and Africa inherited corrupt institutions and thus stayed poor. As economists Daron Acemoglu and Jim Robinson conclude in *Why Nations Fail*, this is why "it is so difficult to remove world inequality and to make poor countries prosperous."[14] But if their conclusion is correct, we should give up on doing development because the past determines the present.

These three schools, each led by towering titans, form the classical literature in political economy. Despite their apparent disagreements, they share one striking commonality: like classical mathematics, they

[8] "Good institutions" refers to the modern institutions found in high-income, industrialized nations, e.g., technocratic government agencies, elaborate laws, state enforcement power, private property rights, formal checks and balances, electoral democracy, and so on.

[9] Inglehart and Welzel 2005.

[10] Sachs 2005.

[11] Munk 2013.

[12] Knack and Keefer 1995. The empirical literature draws inspiration from the influential work of North and Weingast 1989.

[13] Pritchett and Woolcock 2004, 193.

[14] Acemoglu and Robinson 2012, 44.

all make straight-line arguments. Causality points in one direction, from either growth to good institutions, or vice versa, or from history to good institutions to growth.

Despite a dazzling concentration of intellectual prowess, all three schools failed to find a way out of the chicken-and-egg problem. Why? Because they took an inherently endogenous (bicausal) process—economic and institutional development—and forced it to fit within their linear, static models. They did not design a mode of analysis suited to the defining qualities of the subject.

Classical mathematics assumes that the world is made up of regular shapes and as such, it is only well-suited to study the industrial world. By the same token, classical political economy theorizes social systems as mechanical entities made up of disparate parts, with causality pointing in one direction. Development projects are designed around "logframes," a linear matrix of inputs and outputs, like a factory's conveyor belt. Layered on top of an industrial logic is a colonial worldview that assumes that the best, growth-promoting institutions are those found in the West, and backward societies must catch up, converge, and become like Denmark.

Fractal geometry embraces the reality of irregular, messy shapes and uncovers their hidden order. It is well-suited not only to studying the natural world, but also to discovering solutions for the postindustrial age through a more sophisticated and humbler understanding of nature (for example, medical scientists apply fractals to distinguish between normal cells that grow in fractal patterns and cancerous cells that grow abnormally). Similarly, we need a new political economy—one that does not impose artificial assumptions of mechanical properties on complex adaptive social systems. I call this new paradigm *adaptive political economy*.

The Complex Nature of Social Systems

What is the fundamental difference between classical and adaptive political economy? My point of departure begins with clarifying the nature of social systems: international relations, governments, economies, societies, cultures, organizations, institutions, development, innovation, conflict, and so on. Whereas classical political economy views social systems as complicated—a term conflated with complex or messy—adaptive political economy emphasizes that they are complex but not complicated. The difference is not semantic but substantive.

In both popular and social scientific discourse, typically no difference exists between complex and complicated. Analysts use both words as

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synonyms for situations that are hard to grasp and control and that therefore obstruct social scientific inquiry. A best case in point is the classic methodology textbook, *Designing Social Inquiry*, by Gary King, Robert Keohane, and Sidney Verba, whose influence is so profound that followers describe the book in religious terms as "the canonical text of the orthodox camp" and "the complete and unquestionable truth from high."[15] In a section titled, "Science and Complexity," the authors define complexity in this way:[16]

> Social science constitutes an attempt to make sense of social situations that we perceive as more or less complex . . . On the contrary, the perceived complexity of a situation depends in part on how well we can simplify reality, and our capacity to simplify depends on whether we can specify outcomes and explanatory variables in a coherent way . . . Thus "complexity" is partly conditional on the state of our theory.

Notice that the authors understand complexity as a perception, not a reality. More specifically, they see the existence of complexity as a result of scientific deficiency: if only we could simplify more, they believe, we can reduce or remove complexity entirely. Their assumption is echoed everywhere in political science (and increasingly so, as political scientists envy economists and strive to reduce complexity into manageable, small parts). For example, Stephen Haggard states as a matter of fact in his primer on development, "Complexity is not particularly gratifying."[17] (To him, complex means messy.) Another political scientist was aghast when I told her that I was studying development as a complex adaptive process. With the kind intention of saving me from falling off a cliff, she stated, "Politics is *not* the study of the complex."

These assumptions in social science extend, with much higher material stakes, to the corporate world, with companies promising to eliminate complexity from our lives. One advertisement by German-based SAP, a software company specializing in streamlining operations, declares in an advertisement, "Complexity wastes up to 40% of your people's work week." How dreadful!

In fact, complex and complicated are two qualitatively different categories. Complexity is not a perception, but a reality. Some matters objectively possess complex properties that require a different mode of analysis (rough shapes, for example). Important social systems and outcomes invariably fall into this category. Furthermore, complexity is

[15] These quotes appeared in Brady, Collier, and Seawright 2010, 3.
[16] King, Keohane, and Verba 1994, 6.
[17] Haggard 2018, 91.

not a nuisance. Despite appearing messy, complex systems often display intelligent patterns, as fractal geometry demonstrates.

In other words, politics is the study of the complex. If not, then what is being studied is either not important or particularly interesting.

Trees Are Not Toasters

Complex does not mean complicated, just as trees are not toasters. Complicated things are made of many separate parts that do not adapt to one another or the surrounding environment—machines are good examples.[18] A toaster is a machine that is exactly the sum of its parts. To assemble a toaster, follow the instruction manual and put the parts in order (unlike Michelangelo's famous depiction in *The Creation of Adam*, a final magic touch is not needed to bring it to life). When dealing with machines, processes are linear and outcomes can be controlled. Press a button and a toaster will produce a predictable action: crisp, warm bread pops up.

Complex is the opposite of complicated. Whereas machines are complicated, systems are complex. A system is made up of interconnected elements that adapt to one another and the environment, for example, a tree. The term adapt is not just a fancy word for change. Rather, it is a particular type of change, expressed by John Holland as the process by which an agent "fits itself to its environment."[19] This process has at least three key mechanisms: variation (generating alternatives); selection (choosing among or assembling alternatives to form new combinations); and retention (keeping and diffusing a given solution or exploring new ones). Many adaptive iterations result in evolution—that is, substantial changes in a given system.[20]

The differences between complicated machines and complex systems are not semantic but have profound implications for the way social scientists understand causality, indeterminacy, human agency, and institutional design (see Table1).

[18] Here, I refer to simple machines. With artificial intelligence, humans are inventing smarter machines that can learn and adapt to users.

[19] Holland 1996, 9.

[20] Contrary to the concepts offered by James Mahoney and Kathleen Thelen (2010), "institutional evolution" does not equate to "gradual institutional change." Gradual change may not involve adaptation. Wear and tear (in machines) and aging (in humans) both occur gradually, but they are not the result of adaptation. Nor are evolutionary processes always slow moving; microcosms can evolve within hours.

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Table 1
Complicated Does Not Equal Complex

	Complicated	*Complex*
Defining properties	a complicated machine is made up of separate parts that do not adapt to one another or the environment.	a complex system is made up of interconnected parts that constantly adapt to one another and the environment.
Causality	dependent (outcome) versus independent factor (cause)	interdependent factors (both cause and outcome)
Indeterminacy	risk, probability	uncertainty, possibility
Human agency	control	adapt, experiment, learn, influence
Institutional design	institutions (solutions to particular problems)	meta-institutions (systems for enabling the discovery of solutions)
Examples	toasters	trees, ecologies, villages, cities, governments, markets

Source: Adapted from Ang 2016, 52.

Causality: Whereas causal effects in machines are linear, we frequently observe interdependence (or endogeneity) in systems: the cause (independent variable) is also the outcome (dependent variable).[21]

Indeterminacy: We face risk in complicated worlds but uncertainty in complex ones. Mechanical situations pose risks—that is, the probability that certain anticipated outcomes may occur (when a toaster breaks down, a user finds it annoying but not surprising). By contrast, complex systems are constantly adapting and thus generate uncertainty—that is, possibilities beyond anticipation and planning. Some possibilities are terrible, such as financial crises and pandemics. Yet some possibilities, such as groundbreaking innovations or stumbling into one's passion, are marvelous. To extinguish uncertainty is to extinguish possibilities, both terrible and marvelous.

Risks may be predicted and controlled, but possibilities may only be imagined. Confronting risks in complicated worlds, humans seek to exert control to minimize the probability of undesired outcomes. But in the face of uncertainty in complex systems, control may be futile or even self-defeating because in efforts to reach a preplanned outcome, one may be missing better possibilities and paths to those possibilities.

[21] Taking interdependence as a starting point rather than denying it changes our conclusions about how the process of development begins and evolves (see application section).

Human agency: Hence, rather than exerting control to achieve preselected outcomes, the wiser course of action in complex systems is to influence the process of adaptation and learning. A liberal arts education strives to do just this: to teach students how to think, rather than what to think.

Institutional design: Appreciating the distinction between control and influence opens new horizons in institutional design. Political economists have long studied institutions as solutions to particular problems, for example, contract enforcement. But they have not considered a bigger question: Why do some communities devise better solutions than others? In other words, why are some more adaptive than others? These questions point us toward meta-institutions: higher-order structures and strategies that facilitate adaptive and learning processes.[22] Political economists traditionally, however, study control but not learning.

Simple But Not Any Simpler

Given that complicated and complex define two different states, where do social systems fit? Clearly, social systems are complex in nature, resembling trees more than toasters. Numerous actors within social systems, even in a small village, are constantly interacting and adapting to their environment and to one another. Causal relations are interdependent, generating possibilities, both good and bad, and not only risks.

To be sure, complicated and complex are ideal types with varieties in between. Some social scenarios or questions are more mechanical in that they can be reduced to precisely measured variables with a linear effect. These are suitable for experimental designs. For example, does signaling one's party affiliation affect whether drivers will overcharge a passenger?

But the big, enduring themes that matter in social science—modernization, state-building, innovation, conflict, financial crashes, to name a few—are necessarily complex in nature. They feature multiple variables that interact with one another, and the relevant causes may even evolve over time.[23] Causality goes in two directions and effects can be lagged. If we insist that causal inference and experiments are the gold standard and the only legitimate method of analysis, then such a doctrinal view

[22] Adaptive political economy does not preclude quantitative methods. The study of meta-institutions can entail data collection and statistical tests. One example is my project measuring the mixture of clear and ambiguous guidelines in the Chinese bureaucracy, which I argue are a mechanism of adaptive policy communication; Ang 2024.

[23] Ang 2016, chp 5; Ang 2017.

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will preclude the study of questions that matter greatly. Like the proverbial tale of the drunkard who searches for keys under the lamppost because there is light, social scientists are increasingly driven to choose narrow, even trivial, questions that can be precisely answered, while complex themes that do not fit dominant methods are abandoned.

This makes no sense. The keys—the important, tough questions—do not lie under the lamppost. They lie in the dark. As social scientists, our job should be to shed light on the dark. We should be devising appropriate concepts, methods, and theories to illuminate patterns in complex social systems, rather than forcing them to fit mechanical models or ignoring them.

Distilled to a maxim adapted from a quote by Albert Einstein, adaptive political economy strives for *simple but not any simpler*. "Simple" means removing extraneous details when modeling a complex social process. "Not any simpler" emphasizes that we must not, however, distort or ignore a complex subject's defining qualities: interdependence, uncertainty, adaptation, learning, and influence. This understanding opens many new agendas. I provide an example below.

A Demonstration: Development as a Coevolutionary Process

To briefly demonstrate how we can apply adaptive political economy and the fresh conclusions it yields, I refer to my earlier work on development as a coevolutionary process.[24] By nature, economic and institutional development—modernization—is an endogenous (mutually causal) process of change. Yet traditionally, economists respond to this reality by removing endogenous effects using statistical techniques; the jargon "treating endogeneity" indicates that mutual causation is like a disease that has to be cured.

In *How China Escaped the Poverty Trap*, I take endogeneity (coevolution or mutual change) as my starting point, as a natural process to be understood rather than purged from analysis. To this end, I developed a qualitative method for tracing and modeling its patterns. First, I introduced a template for mapping coevolution in any two given domains (see Figure 1), adapted from earlier work by Johann Murmann on coevolutionary processes in the business sector.[25] I applied this framework

[24] Ang 2016, chp 5; Ang 2017.
[25] Murmann 2013.

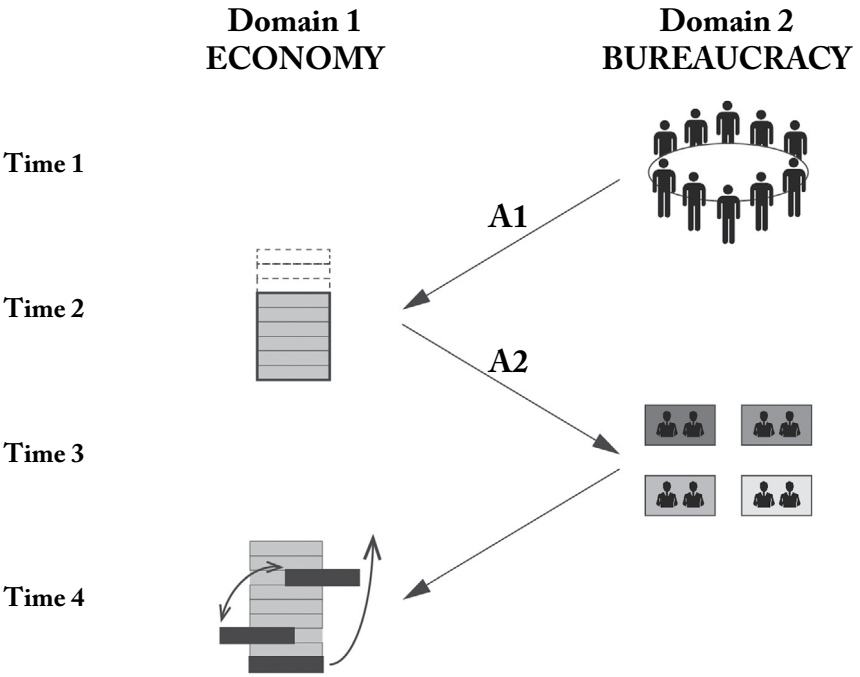

Figure 1
Four Steps of Mapping Coevolution Paths

Source: Reprinted from Ang 2017.

to map the coevolution of states and markets, and more specifically, that of the economy, bureaucratic goals and incentives, and diverse forms of property rights. I generated many detailed coevolutionary maps, with an example shown in Figure 2.

While my focus was on China's great transformation during the reform and opening period (1980s through 2012), I extended the same approach to three other disparate settings: the expansion of trade in late medieval Europe, the evolution of public finance in the antebellum United States, and the unlikely success of Nollywood in contemporary Nigeria. In other words, applying my framework to China does not limit its relevance to China, although some insist it does.

If we strip away all the particularities, what common pattern do we see? My answer is shown in Figure 3. In its most distilled form, development is best understood as a three-step, coevolutionary process.

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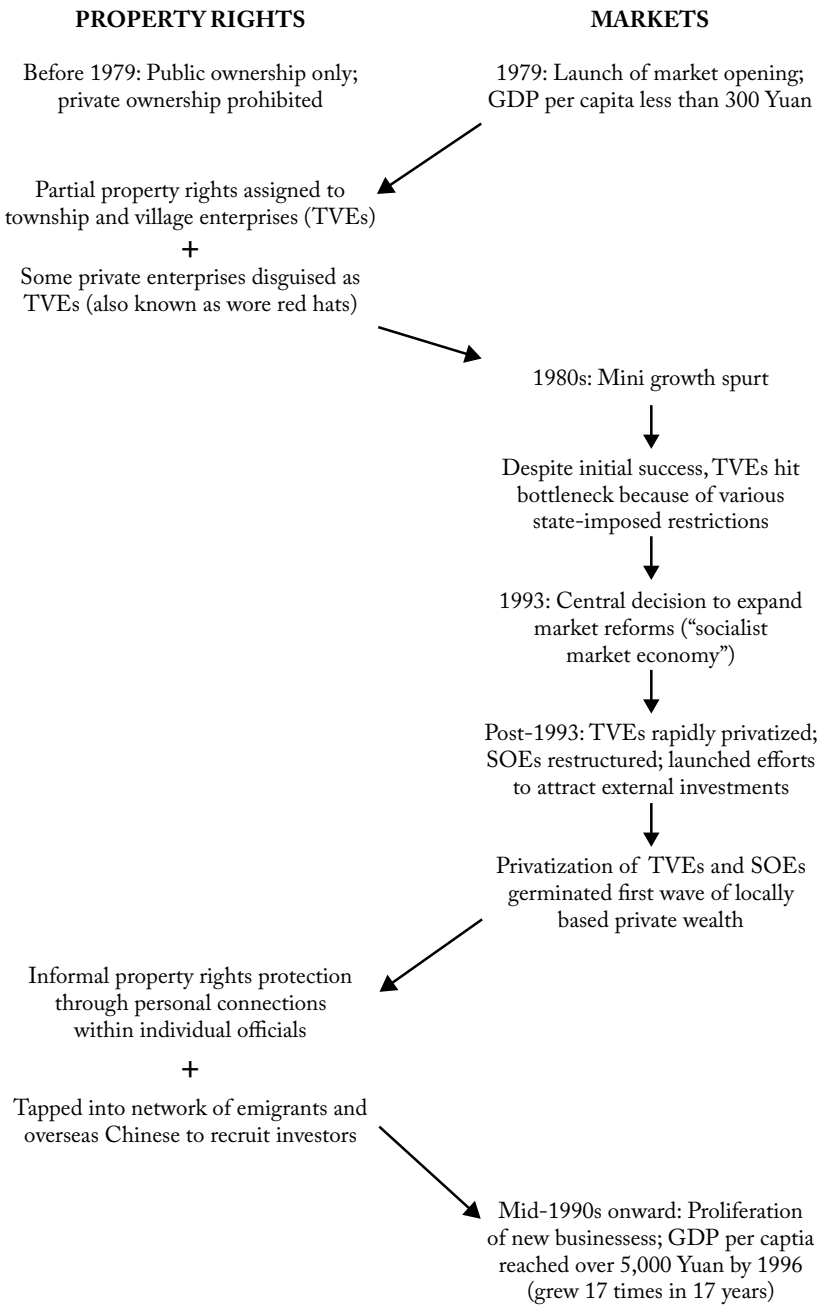

Figure 2
Coevolution of the Economy and Property Rights

Source: Ang 2016, 154.

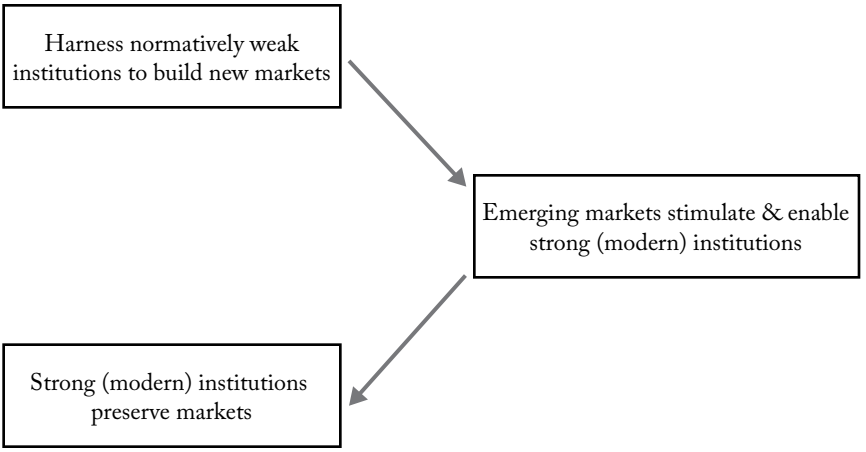

Figure 3
Development as a Three-Step, Coevolutionary Process

Step 1: Harness normatively weak institutions to build new markets.

— 1a. In China, entrepreneurs relied on collective and informal property rights, rather than on formal protection of private property rights, to kick-start economic activities.

— 1b. In Nigeria, entrepreneurs relied on piracy and cheap production of low-quality films, rather than on formal protection of intellectual property rights, to kick-start a new industry.

Step 2: Emerging markets stimulate and enable strong (modern) institutions.

— 2a. In China, as the economy took off, private entrepreneurs grew in size and influence and desired a predictable business environment.

— 2b. In Nigeria, as the industry took off, filmmakers desired to earn more revenue and make higher-quality films.

Step 3: Strong (modern) institutions preserve markets.

— 3a. In China, private entrepreneurs demanded an end to bureaucratic harassment and sought formal participation in political institutions such as legislative councils.

— 3b. In Nigeria, filmmakers, particularly those targeting a higher-end market, pushed for formal intellectual property rights protection.

I highlight two important, fresh conclusions from my alternative model of development.

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First, the institutions, methods, or capacities for building new markets look and function differently from strong (modern) ones that later evolve to sustain mature markets. Indeed, market-building institutions often look wrong to first-world elites. That is why I stress in step one that such institutions are only normatively weak—that is, weak from the first-world perspective.

What do conventional models miss? They incorrectly assume that growth-promoting institutions come in one universal package like those found in idealized Western liberal economies. They fail to distinguish between the different needs, constraints, and resources at early and advanced stages of development and the different solutions that could fit. Without acknowledging so, they judge what is best from the standards of the modern and typically Western.

Second, the first step of development is using what you have, not what you want. The poor cannot innovate with what they lack: wealth and modern capacity. Necessarily, they must creatively repurpose what they have—practices and resources that the rich may dismiss as backward—to kickstart change. A key word here is "repurpose," meaning existing materials do not perform miracles by themselves. For example, in China, township and village governments had long been in place, and, when market reformists prevailed over Maoists, they repurposed these preexisting units to create collective property rights and surrogate entrepreneurs.

Another key word is "innovative actors." Analysts know that innovation is integral to development, but they tend to portray innovation as the exclusive purview of the educated or rich. What my model emphasizes is that all development begins with modest forms of innovation among nonelites, the poor, or grassroots actors.[26] This reality has been all but ignored by conventional models privileging foreign aid, first-world best practices, and good colonial legacies in North America.

In sum, this example demonstrates what I mean by "simple but not any simpler." I reduced a mass of moving parts into its most essential form: development as a process of three coevolutionary steps. This exercise is no mere storytelling; it reaches new conclusions with theoretical (how development begins and evolves), practical (solutions to promote), and moral (actors to empower) implications. If I made my model any simpler, in the convention of two linear steps, it becomes misleading because so doing erases the distinction between market-building and

[26] Ang 2021.

market-preserving stages of development and the different solutions that fit each stage.

Conclusion

Adaptive political economy is not a new theory, but rather a new paradigm. A theory is an explanation or solution for a particular problem. A paradigm is a new way of seeing and studying the social world. It is a mistake to conflate complicated with complex and to apply reductionist methods suited to mechanical objects upon complex, adaptive social systems. Making sense of complex social systems and processes calls for a wholly different approach—simple but not any simpler—just as modeling rough shapes calls for fractal rather than classical geometry.

Fractal geometry, however, is only a partial analogy for the intellectual quest at hand. Changing social paradigms is more difficult than paradigms in the natural science context because social systems are colored by power asymmetry and normative biases. The modern world is not only a product of industrialization but also that of colonialism. As Edward Said reminds us, "No one has ever devised a method for detaching the scholar from . . . a class, a set of beliefs, a social position."[27] This perspective is sorely missing in complexity economics, which does not recognize the power dynamics that determine which ideas are accepted and which ones are not.

Take the political economy of development. Once, when I presented on "using what you have" to an elite department, one academic scoffed and said, "This sounds too simple." That is privilege speaking. It is also double standards on display. Do not the standard prescriptions of foreign aid or replicating rich-nation practices sound too simple? These elites who dismiss innovation among the poor are enthused about designing foreign-imposed aid projects to save the poor.

Adaptive political economy, therefore, must incorporate an inclusive and moral dimension—making it, collectively, an adaptive, inclusive, and moral (AIM) political economy—that honestly recognizes the inequities that have molded the way we think, see, and approach the world.

Many experts and policymakers see the twenty-first century as a dreadful time of polycrisis, when all the excesses of modernization have come home to roost and manifest physically in a climate crisis. But it is

[27] Said 1978. See also Shilliam, who reminds us: "Even when the categories do not directly reference colonialism, a colonial logic can still operationalize them . . . If comparative methods cannot pursue these reflections, it will remain as politically conservative as it was during the days of empire and the era of decolonization"; Shilliam 2021, 117.

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also a precious opportunity to reflect on the limitations of the existing paradigm that sprung out of an industrial-colonial age and on the possibilities for change.

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Key Words

complexity, complex adaptive systems, political economy of development, endogeneity, coevolution, market-building, market-sustaining, adaptive, inclusive, moral